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of the European Union

"Sustainable Blue Economy"

DiveSafe

"Integrated system for scientific and environmental underwater surveys, with advanced health & safety features"

DIVE SAFE

WP5: Commercialization, Sustainability & Market-readiness (TRL8)

Software Requirements Specification – App for USBL bundle interfacing

Responsible Organization: UnivPM

Version 0.1, Date: 21/12/2020

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Preface

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Date	Version	History of change
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1. Introduction

1.1 Purpose

This document presents the software app for USBL bundle interfacing.

1.2 Intended Audience and Reading Suggestions

This document is intended for developers and users who need to know and use the DiveSafe USBL bundle interface system. Therefore, it starts from the product perspective, including all DiveSafe system to detail the app for USBL bundle interfacing.

1.3 Product Scope

The purpose of this software is to act as a communication bridge between the Mission Manager (the software with a graphic interface that allows the surface user to view the mission and interact with the divers) and the USBL transceiver, which will be hung under the boat or a buoy. In particular, this means that every message transmitted from the depth to the surface and vice versa must pass and be interpreted and encrypted by this software according to the defined protocols.

1.4 References

- DiveSafe Deliverable D3.1
- DiveSafe Deliverable D3.5
- DiveSafe Deliverable D3.6

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2. Overall Description

2.1 Product Perspective

The DiveSafe system (Figure 1) is composed of several modules:

- The **underwater module** that includes:
  - an underwater scooter equipped with a Docking Station (DS) able to acquire and exchange data coming from several environmental sensors, health sensors, the underwater tablet and the acoustic localization system that are connected to it;
  - an Underwater Tablet (UT) running the Navigation App that supports the diver during the immersion providing several functions, such as assisted navigation, messaging features, health sensors monitoring, possibility to take geo-referenced photos, etc.;
  - a high-resolution camera;
  - environmental sensors payload (i.e. temperature and pressure);
  - wearable sensors for breathing/heartbeat and glycemia monitoring;
  - an acoustic modem to support both USBL localization and communication.
- A **topside unit** that includes:
  - a personal computer or a laptop running the Mission Manager app that enables the surface operator to plan a mission, monitor the divers' position and health status during the underwater operations and to communicate with them;
- A **cloud module** that is composed of:
  - a Remote Server (RS) that hosts the database to store all mission plans and data gathered during the missions and enables other modules to interact with the database through a Web Services API (WS API);
  - a 3D reconstruction module that can perform, within one day, the 3D reconstruction of the underwater site based on photogrammetric documentation.

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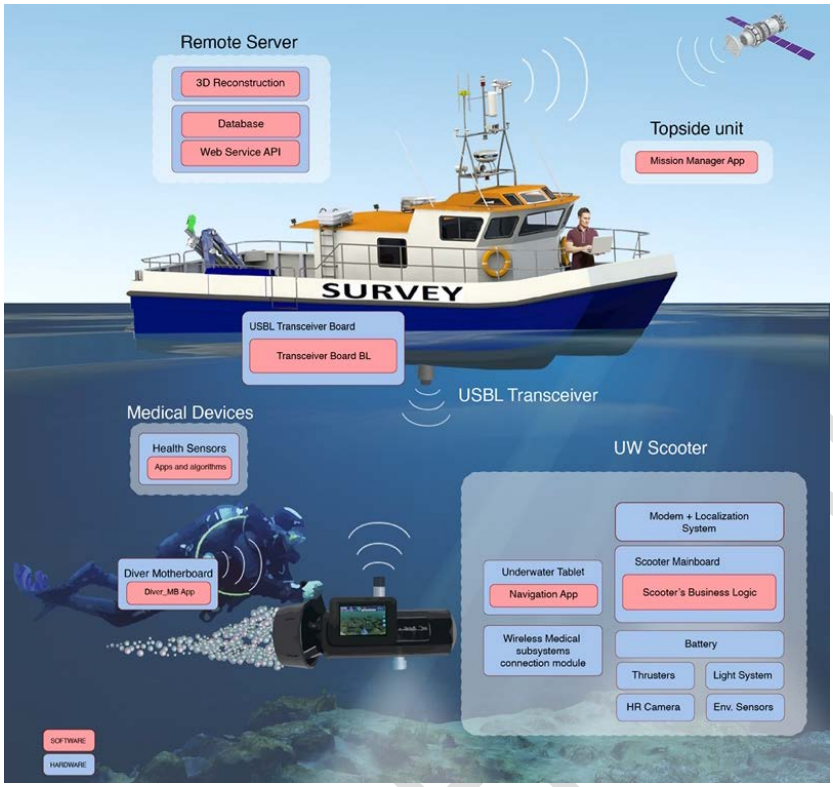

Figure 1. DiveSafe HW and SW architecture

As shown in Figure 2, the **underwater module** is composed of various components. The tablet is placed in an underwater case that ensures access to its features through a touch pen. It is anchored on the scooter or DS through the cradle. The DS is provided with an embedded mainboard that manages all the electronics and implements all the business logic. The communication between the DS and the UT is done through a Wi-Fi channel. The Diver_MB is composed of an embedded board that is placed on the diver suite. It enables physiological data logging from wearable sensors and their transmission to the scooter. The acoustic modem is anchored to the scooter.

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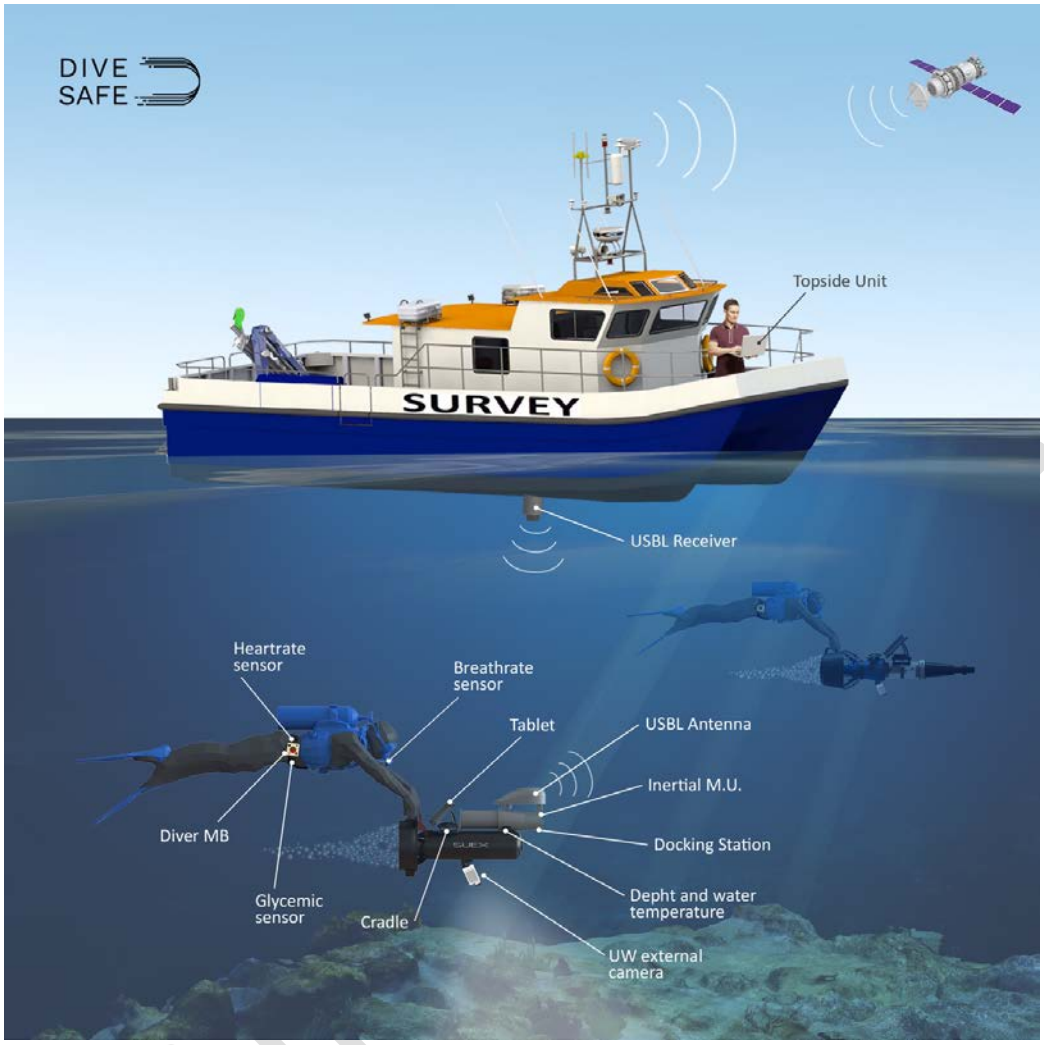

Figure 2. DiveSafe Underwater module components

Through the Mission Manager application (MM), the **topside unit** enables the surface operators to plan new missions by placing some targets on top of the 3D model of the underwater site. The mission plan will be stored in the RS and then will be download by the UT just before starting the underwater activities. Then, surface operators can monitor the current activities of divers and their health conditions and communicate with them.

The Cloud Module is composed of a Remote Server system that enables to store all planned missions and data gathered by divers during the underwater activities. The storage is made through a central database and a file system (mainly used for photos and 3D models). The access to this data is managed by a RESTful WS API and JSON data-interchange formats. Moreover, the Cloud Module includes the module in charge of managing the 3D reconstruction of the seabed, starting from divers' photos. Finally, the RS for data managed can be deployed as a Local Module in order to be used on the surface boat without an internet connection.

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2.2 Product Functions

This product acts as follows:

- Gets the position from the GPS sensor
- Sends the message to the USBL transceiver to the head to locate the scooter in the water. If successful, transmit the data to both the MM and the scooter
- Receives commands from the MM with consequent Sending of start and stop heal messages or chat messages to the scooter via USBL transceiver
- Receives all messages transmitted, including those from Heal and send them to the MM.

2.3 User Classes and Characteristics

The user who will use this app must have no particular knowledge other than that of the system as a whole and the interface of the MM in detail.

2.4 Operating Environment

The software written in python was then converted to .exe so it must be launched from a windows 10 64 bit PC.

The laptop on which this software runs must necessarily be on a boat during sea operation, or on a shore of a surface during a lake or swimming pool operations.

2.5 Design and Implementation Constraints

This software is subject to two types of constraints:

- The messages transmitted to the MM must have a minimum interval between them of at least 0.2 s because the MM can manage messages that arrive with a shorter interval if a message arrives when the MM has not yet finished managing the previous one it interrupts the first and passes to the second
- The most important constraint is due to the transmission times of the acoustic wave in the water. In particular, the time to locate a device is 1-1.5 seconds per device, provided that the first attempt is successful. This means that no other messages can be sent in this time interval and no other devices can be located. So in the case of two devices, the localization frequency per device will be about 0.2 Hz.

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3. External Interface Requirements

3.1 User Interfaces

The software is launched from the command line, and debug messages are displayed. The command is:

Main_PC.exe [TCP Port] [UDP Port]

The meaning of the ports will be specified in the next section.

The user cannot interact with the software in any way except to close or open it.

3.2 Hardware Interfaces

The laptop running the software must have 2 connections with other hardware:

1) Telnet connection via ethernet cable or Wi-Fi network with the USBL transceiver. Wi-Fi is required if the head is attached to the surface buoy.
2) Connection via serial port to a GPS to know the position of the USBL transceiver. All GPS that connect via serial and use NMEA protocols are supported.

3.3 Software Interfaces

Dummy 1 is the application developed to manage the USBL Transceiver located on a buoy or under the boat; it receives the command from the Mission Manager (MM) and sends data. The communication protocol API defines the set of messages between the two systems; depending on the message type, one of the two standards TCP/UDP communication protocols will be used, in particular:

- TCP channel is reserved for command messages. This class of messages includes start and stop messages of the tracking system, messages for scooters, etc.
- UDP channel is reserved for real-time data transmissions such as diver location and health data from the wearable sensors

Data exchange procedure between Dummy 1 and Mission Manager is displayed in Figure 3.

Dummy 1 is connected via telnet to the USBL Transceiver.

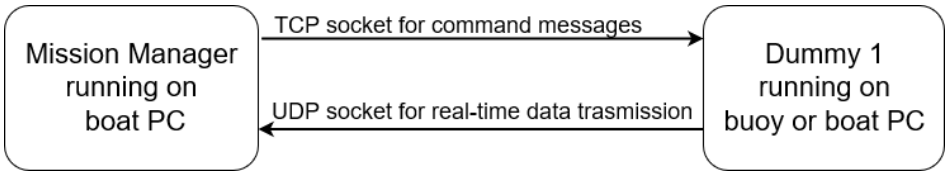

Figure 3: Data exchange between MM and Dummy 1

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3.3.1 Command message format

The structure of the commands between MM and Dummy 1 over TCP and between DS and the sensors is:

Header (2Bytes) | Type|Vendor | Model|Address | CSR Address | DataLength | Header Checksum | Payload Data | Total Checksum

Bytes: |2|1|1|1|1|1|max 254|1 = [8..262]

Header: Request - 0xFAAF ; Response - 0xFDDF

Type|Vendor: in order to send the data only to a device of specific type and vendor.

The 0x0F value is used for a broadcast message (i.e. 0xyF as Broadcast to all devices of *y* type)

Model|Address: To send the data only to a device of the specific model and with a specific address.

CSR Address: Constant byte set to 0x00.

DataLength: This is the length of the payload data. The count of bytes between the Header Checksum and the total checksum.

Header Checksum: checksum of all the bytes from the first start byte to the DataLength byte included.

Payload data: This is either the data to be written to the device or data sent as a response.

Request: **Data[0]**: command

Data[1]: data length of the eventual data for the command

Data[2]: eventual data for the command

Answer: **Data[0:2]**: Original Command - command of which this is an answer

Data[3:4]: Answer data first 2 bytes - operation result (successful or not)

Data[5]: Answer data third byte - data length of the eventual answer content

Data[6+Data[5]]: Answer data content

Total Checksum: This is the bitwise XOR of ALL bytes from the first start byte to the last payload data byte included. Note: This is also just the Data Payload checksum since the first 6 bytes of the header XOR with the header checksum is 0.

Commands messages are messages from the UT for the DS addressing specific actions. At least the following Type, Vendor, Model and Commands are required:

Type:

0: Docking Station Jetson

1: USBL **Vendor**: 0: Sonardyne **Model**: 0: Scout

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1: EvoLogics **Model**: 0: [S2C-524-05-R] S2C M 18/34 Hydroacoustic Mini Modem + Streamlined Head for USBL

1: EvoLogics **Model**: 1: [S2C-604-01-E-C] S2C R 18/34 Hydroacoustic USBL + Streamlined Head for USBL

1: EvoLogics **Model**: 2: [S2C-604-01-E-x] S2C R 18/34 Hydroacoustic USBL

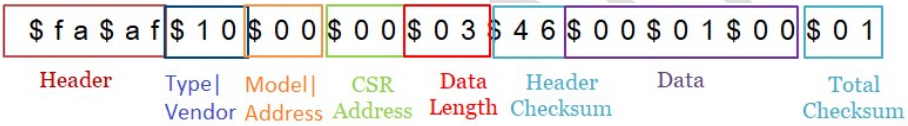
2: Blueprint **Model**: 0: Seatrac

2: Light

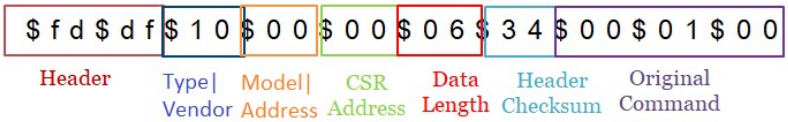
3: Battery

4: if the message is for an acoustic modem installed on a scooter, the **address** field is the address of the scooter which can be a number ranging from 1 to 8.

Request:



Answer:



Commands for USBL:

- 0x00: STOP_LOCALIZATION
- 0x01: START_LOCALIZATION: start the localization

  This command must be followed by the number of scooters (payload length) with the respective addresses (the address can be a number from 0 to 8) For example, if I want localize two scooters with address 3 and 4 the message will be: fa:af:00:01:00:04:50:01:02:03:04:04

- 0x02 SEND_message to a scooter

  For this message the type must be 4 and vendor the scooter address. The payload is composed by the string message (MAX 64 characters)

- 0x03 START_HEAL MESSAGE

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- 0x04 STOP_HEAL MESSAGE
- 0x05 GET_HEAL_from_Scooter_x: the Payload of the message is the scooter address
- 0x06 GET_position_from_Scooter_x: the Payload of the message is the scooter address

3.3.2 UDP Message

Over UDP, Dummy 1 will send all the information about health. The Following protocols are used:

Positioning Message

Positioning message labels are displayed in Table 1.

Field	Description
#DS_POS	Header, (fixed
hhmmss,xx	Data acquisition time
T	Fixed
n	Scooter address or 0 for the ship position
DD.dddddd,	Latitude in degrees
N,	Direction of latitude (N or S)
DDD.dddddd,	Longitude in degrees
E,	Direction of latitude (E or W)
DD.dd	Depth in meters
M,	Fixed
***hh**	Checksum

Table 1: Positioning message labels

Example: #DS_POS,112233.44,T,3,43.516889,N,013.546783,E,10.33,M*I

Health Message

Health message labels are displayed in Table 2.

Field	Description
#DS_HEAL,	Header, (fixed
hhmmss,xx	Data acquisition time
T	Fixed

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n	Scooter Address
DD.dd,	Glucose in mg/dL
GL,	Fixed
DD	Breath rate in breath/min
BR,	Fixed
***hh**	Checksum

Table 2: Health message labels

Example: #DS_HEAL, 123456.43,T,80,GL,60,BR,*T

CHAT Message From scooter

Chat message labels are displayed in Table 3.

Field	Description
#DS_MES,	Header, (fixed
hhmmss,xx	Data acquisition time
T	Fixed
n	Scooter Address
DATA,	Message
***hh**	Checksum

Table 3: CHAT message labels

For example: #DS_MES,114433.11,T,3,HELLO*T

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4. System Features

4.1 USBL transceiver and GPS position

4.1.1 Description and Priority

The software must acquire the position via serial from a GPS client. The GPS needs an adapter to be connected to the PC. The priority of the task is high.

Once the first GPS position has been acquired, the software starts transmitting it both to the MM and the devices in-depth, particularly for the latter every 15 s not to clog the acoustic line.

4.1.2 Stimulus/Response Sequences

The user simply needs to connect the USB adapter before starting the software

4.1.3 Functional requirements

The GPS must provide the position in NMEA format, especially GPGGA. The software will not start the localization and then the stream towards the MM until it has received the boat's first position.

4.2 Localization

4.2.1 Description and Priority

Scooter localization is an essential task of this software. This happens in 4 steps:

- Sending a message of minimum size to a scooter
- The response of the scooter with an ack. At this point, the USBL transceiver calculates the distance from it in meters and transmits it to the PC
- Transformation of position into GPS coordinates
- Position transmission to the MM and the Scooter

4.2.2 Stimulus / Response Sequences

The task takes place automatically. The user only has to start it by pressing the start button on the MM.

4.3 TCP Command receiver

4.3.1 Description and Priority

The software must receive various commands over TCP according to the protocol (section 3.3)

- Start
- Stop

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- Health data request (broadcast)
- Stop Health data (broadcast)
- Request Greetings Scooter Number
- Request Position Scooter Number

Each request is followed by sending a message to the scooter(s) via the USBL transceiver.

In the case of broadcast messages, the software does not expect a response ack but remains listening to receive the heal messages, while in the other case the software will try to send the message until the ack is received.

4.3.2 Stimulus / Response Sequences

In this case, the user must be able to send commands using the appropriate MM buttons

4.4 Telnet/reader UDP/sender

4.4.1 Description and Priority

Every time the software receives a string via telnet, it parses that the string has a proprietary Evologics protocol on telnet. If the received message contains a position, health or chat data, it is immediately transmitted via UDP via protocol to the MM

4.4.2 Stimulus / Response Sequences

This Feature takes place completely automatically, every time a data arrives, it is transmitted according to the protocol to the MM, which then displays it.

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Appendix A: Glossary

Abbreviation	Explanation
RS	Remote Server
DPV	Diver Propulsion Vehicle
LC_DPV	Low-Cost DPV
HE_DPV	High End DPV
DS	Docking Station
DS_Modem	DS with the Mini-Modem
USBL	Ultra-Short Baseline
DS_USBL	DS with USBL and external antenna
API	Application Programming Interface
Diver_MB	Diver Motherboard
UT	Underwater Tablet
MM	Mission Manager
NA	Navigation App

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